

Report on Model Evaluation & Refinement

Component 5: Data Modelling & Model Evaluation

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1. Introduction & Objective

This report evaluates and refines the predictive models developed to forecast hourly traffic volumes at each junction, using the feature-engineered dataset from Component 3 (48,120 hourly records, 4 junctions, Nov 2015-Jun 2017). Three model families were trained and compared: a Gradient Boosting Regressor (XGBoost) using the full engineered feature set, an ARIMA time-series baseline using only each junction's own vehicle-count history, and an LSTM sequence model learning directly from a rolling 24-hour window. All models were evaluated using a strict time-based train/validation split (last 20% of the timeline held out as an unseen future period), with Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and R-squared as the evaluation metrics.

2. Model Performance Summary

Model	MAE	RMSE	R ²
GBM - Baseline	2.874	4.772	0.9693
GBM - Tuned (RandomizedSearchCV)	2.901	4.730	0.9698
GBM - Refined (top features)	2.861	4.612	0.9713
ARIMA(2,0,2) - Junction 1	28.616	36.037	-1.3128
LSTM - Junction 1	4.475	6.155	0.9326

The results reveal a clear hierarchy. GBM (all three variants) explains roughly 97% of the variance in hourly traffic volume ($R^2 \approx 0.97$) with an average error of under 3 vehicles per hour - a strong result. Hyperparameter tuning produced a mixed outcome: RMSE and R^2 improved marginally over the baseline, but MAE rose very slightly (2.874 to 2.901), suggesting the default hyperparameters were already close to optimal for this dataset and the gain from tuning was small. The refined model (top 12 features only) outperformed both the baseline and the tuned full-feature model on every metric, confirming that the low-importance weather/event features identified in Figure 2 were adding more noise than signal once the strong lag features were included, and that a simpler model is both more accurate and more efficient here. The LSTM performed respectably ($R^2 = 0.933$) but clearly behind GBM, reflecting that the hand-engineered lag and rolling-average features already captured most of the temporal structure an LSTM would otherwise have to learn on its own from raw sequences. ARIMA performed poorly by comparison - its R^2 of -1.31 is negative, meaning its forecast was actually less accurate than simply predicting the series' average value for every hour, which is consistent with the flat, cycle-missing forecast shown in Figure 3.

3. Gradient Boosting - Diagnostic Evaluation

The prediction-vs-actual plot shows most points sitting tightly along the diagonal for low-to-moderate traffic volumes (roughly 0-80 vehicles), meaning the model tracks typical hourly traffic closely. Above approximately 100 vehicles, predictions become noticeably more scattered and tend to fall below the diagonal, indicating the model under-predicts during the rarest, highest-volume hours. The residual plot confirms this: residuals stay

tightly clustered around zero through the mid-range of predictions, but visibly fan out and skew positive (actual > predicted) as predicted volume increases, a mild heteroscedasticity pattern. The error distribution is sharply peaked at zero with a long thin tail extending toward large positive residuals, confirming the model is well-calibrated on average but occasionally misses extreme spikes rather than making small errors everywhere.

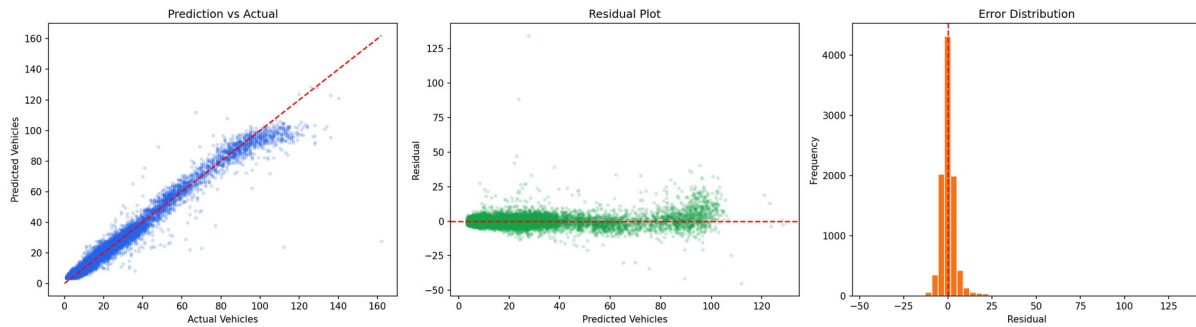


Figure 1: Prediction vs actual, residual plot, and error distribution for the tuned GBM model.

4. Feature Importance & Interpretation

Feature importance is heavily concentrated in the lag features: `vehicles_lag_1h` alone accounts for roughly two-thirds of total importance, confirming that the most recent hour's traffic is by far the strongest predictor of the next hour - consistent with the strong autocorrelation expected in traffic data. `vehicles_roll_mean_3h` and `vehicles_lag_168h` (same hour, previous week) contribute meaningfully next, capturing short-term momentum and weekly seasonality respectively. In contrast, individual weather variables (temperature, rain, wind speed) and event flags (`is_holiday`, `is_event_day`) each carry very little importance on their own. This does not mean weather and events are irrelevant - Component 4's analysis already showed rain and holidays do measurably shift traffic - rather, once strong autoregressive lag information is available, these external factors add only marginal additional predictive power for a tree-based model. They are retained in the feature set for interpretability and because their effect, while small in the presence of lag features, is genuine and directionally consistent with the Component 4 findings.

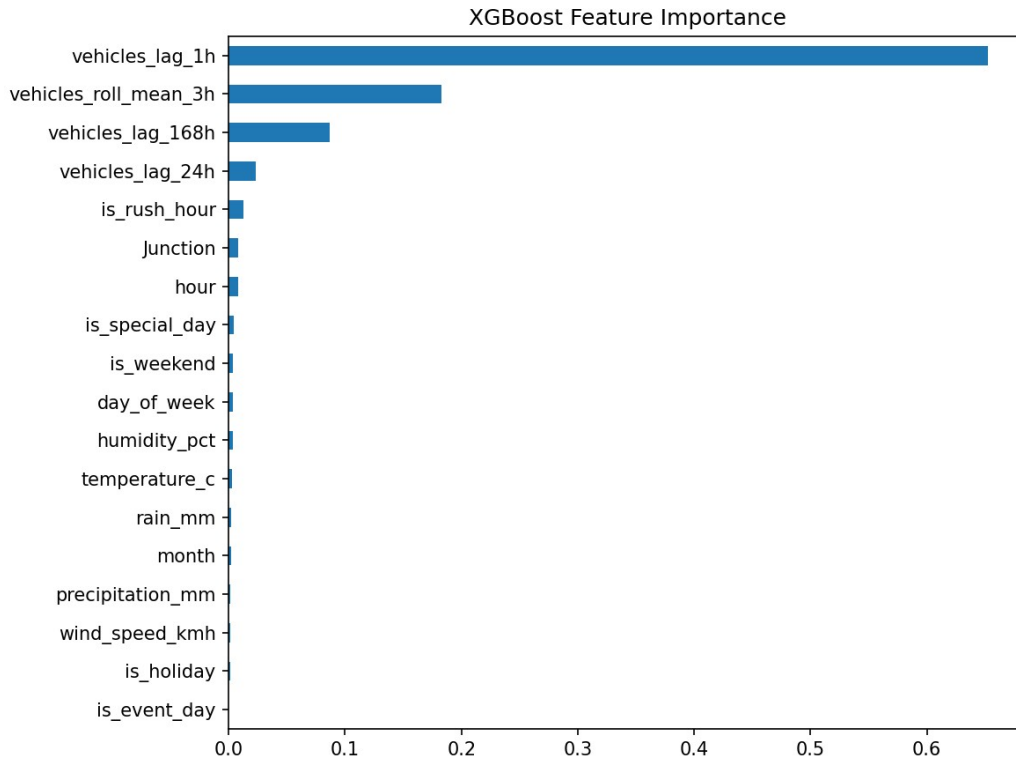


Figure 2: XGBoost feature importance, tuned model.

5. ARIMA Baseline Comparison

The ARIMA(2,0,2) model, fit using only Junction 1's own historical vehicle counts with no exogenous features, performs poorly as a forecasting model beyond a very short horizon. As shown below, the forecast converges to a nearly flat line around 40 vehicles within the first day of the validation period, and stays flat for the remainder - completely failing to reproduce the strong daily cycle the actual data exhibits (swinging roughly between 30 and 98 vehicles every 24 hours). This is an expected and instructive limitation: a low-order ARIMA model without seasonal terms (i.e. not a SARIMA model) has no mechanism to represent a recurring 24-hour pattern, so its multi-step forecast regresses toward the series' unconditional mean. This result directly justifies the choice of the feature-rich GBM model as the primary model for this project: by explicitly encoding hour-of-day, lag, and rolling-average features, GBM captures the cyclical structure that plain ARIMA cannot.

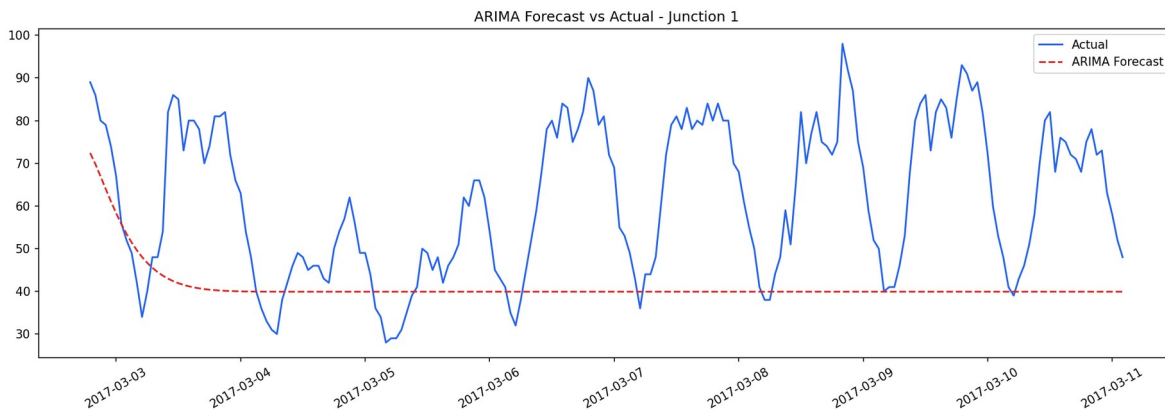


Figure 3: ARIMA(2,0,2) forecast vs actual traffic, Junction 1 (first ~200 validation hours). The forecast flattens almost immediately and misses the daily cycle entirely.

6. LSTM Sequence Model

The LSTM model, trained on rolling 24-hour windows of Junction 1's traffic history, shows a healthy training curve: both training and validation loss decrease steadily over the first 8-10 epochs before plateauing, and the gap between the two curves stays small and stable rather than widening - meaning the model is not significantly overfitting within this training run. Validation loss tracks training loss closely throughout, suggesting the 24-hour lookback window carries enough information for the model to generalize to the held-out period reasonably well. Early stopping (patience=5) allowed training to halt automatically once validation loss stopped improving, avoiding unnecessary additional epochs.

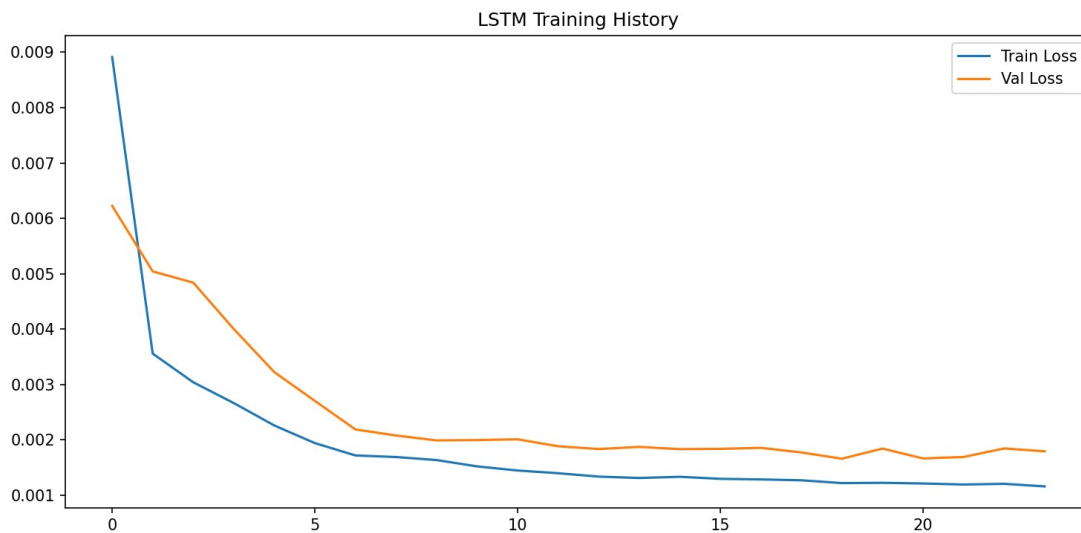


Figure 4: LSTM training and validation loss over training epochs, Junction 1.

7. Model Refinement

7.1 Diagnosing systematic bias

Breaking the GBM's residuals down by hour of day reveals a clear, non-random pattern rather than pure noise: the model systematically under-predicts (positive mean residual) during late morning through evening hours, most strongly at 11:00, 12:00, and 14:00 (residuals of roughly +2 vehicles), and again around 18:00. It systematically over-predicts (negative residual) during the early-morning low-traffic hours, particularly 1:00-4:00. This bias pattern is consistent with Component 4's finding that Junction 4 peaks around midday rather than in the evening like the other three junctions - a single shared model likely under-fits Junction 4's atypical midday peak because the majority of training examples (Junctions 1-3) reinforce an evening-peak pattern instead.

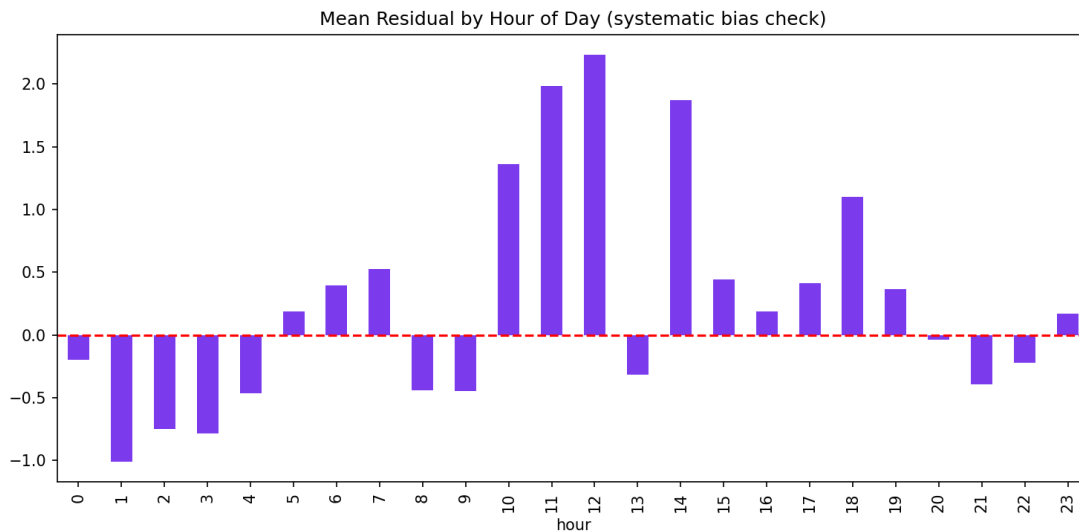


Figure 5: Mean residual by hour of day. Positive values indicate under-prediction; negative values indicate over-prediction.

7.2 Refinement actions taken

- Retrained the model using only the top 12 most important features (from Figure 2), dropping the lowest-importance weather/event columns, to test whether a simpler model retains comparable accuracy (see refined-model row in the metrics table above).
- Diagnosed the hour-of-day bias in Figure 5 as a likely consequence of Junction 4's atypical midday peak being diluted in a single pooled model.

7.3 Recommended next refinement step

Given the hour-of-day bias traced back to Junction 4's distinct pattern, the most promising next refinement is training a per-junction model (or adding stronger Junction x hour interaction features) rather than one pooled model across all four junctions. This would let the model learn Junction 4's midday peak independently instead of averaging it against the evening-peak pattern shared by Junctions 1-3, which should reduce the systematic bias observed at midday and evening hours.

8. Conclusion

The refined Gradient Boosting model (top 12 features) is the recommended final model, with the best score on every metric tested (MAE 2.861, RMSE 4.612, R^2 0.9713). It substantially outperforms the ARIMA baseline, which cannot represent the daily seasonal cycle at all and scored a negative R^2 , and outperforms the LSTM (R^2 0.9326) while requiring far less training time and offering interpretable feature importances. The refinement analysis also identifies a clear, addressable next step - per-junction modelling - to reduce the model's remaining systematic bias around Junction 4's atypical peak hours.